

## Reflexive

### § Reflexive

#### Parameters

- +  $A \in \text{Set}$
- +  $R \in A \sim A$

#### Assumptions

- a.  $\text{id}_A \subseteq R$

#### Observations

1.  $(\forall a \mid a \in A : a \text{ R } a)$
2.  $(\forall a \mid a \in X : (\exists x \mid x \in X : a \text{ R } x)) \Leftarrow X \subseteq A$

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### Proof of Observation 1

◆  $\text{id}_A \subseteq R$

$\equiv$  Extensionality

$(\forall y, x \mid y \text{ id}_A x : y \text{ R } x)$

$\equiv$  Definition of identity

$(\forall y, x \mid y = x \wedge x \in A : y \text{ R } x)$

$\equiv$  One-point rule

$(\forall x \mid x \in A : x \text{ R } x)$

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### Proof of Observation 1

◆  $(\forall a \mid a \in X : (\exists x \mid x \in X : a \mathbf{R} x))$

i.  $R \in A \sim A$

ii.  $(\forall x \mid x \in X : x \in A)$

iii.  $(\forall a \mid a \in A : a \mathbf{R} a)$

⊢ **Generalisation, fixing  $a \in X$**

◆  $(\exists x \mid x \in X : a \mathbf{R} x)$

iv.  $a \in X$

1.  $a \in A$

⇐ **∃ introduction**

$(x \in X \wedge a \mathbf{R} x)[x := a]$

≡ **Textual substitution**

$a \in X \wedge a \mathbf{R} a$

≡ **Assumption iv.**

$a \mathbf{R} a$

≡ **Assumption iii. using observation 1.**

true

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